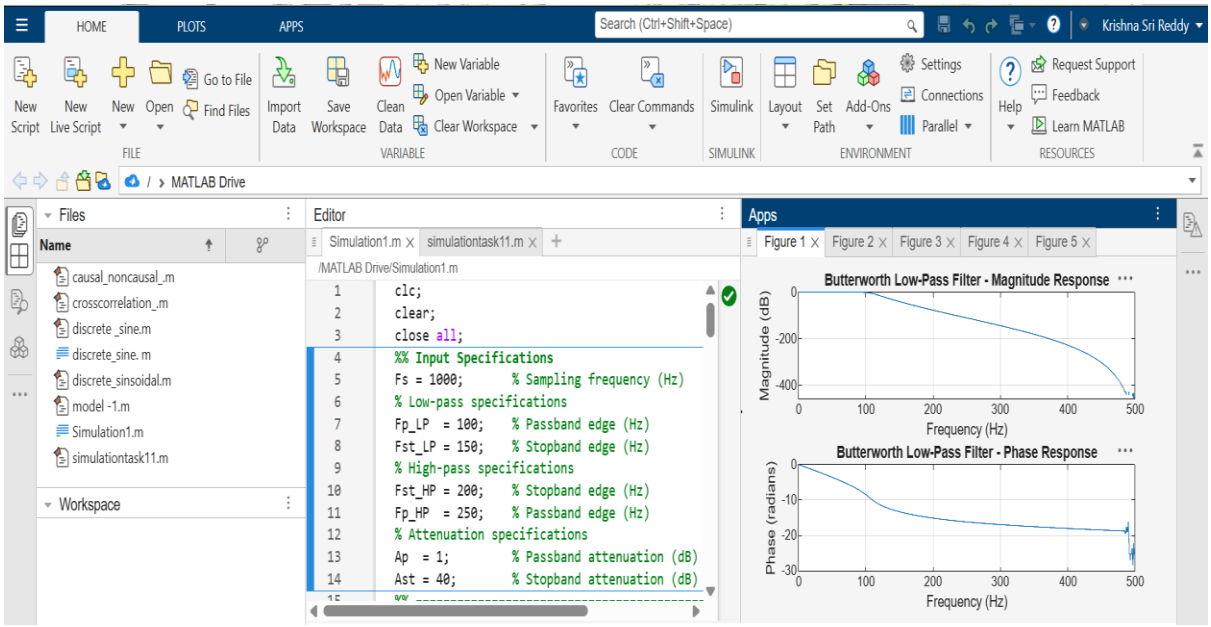
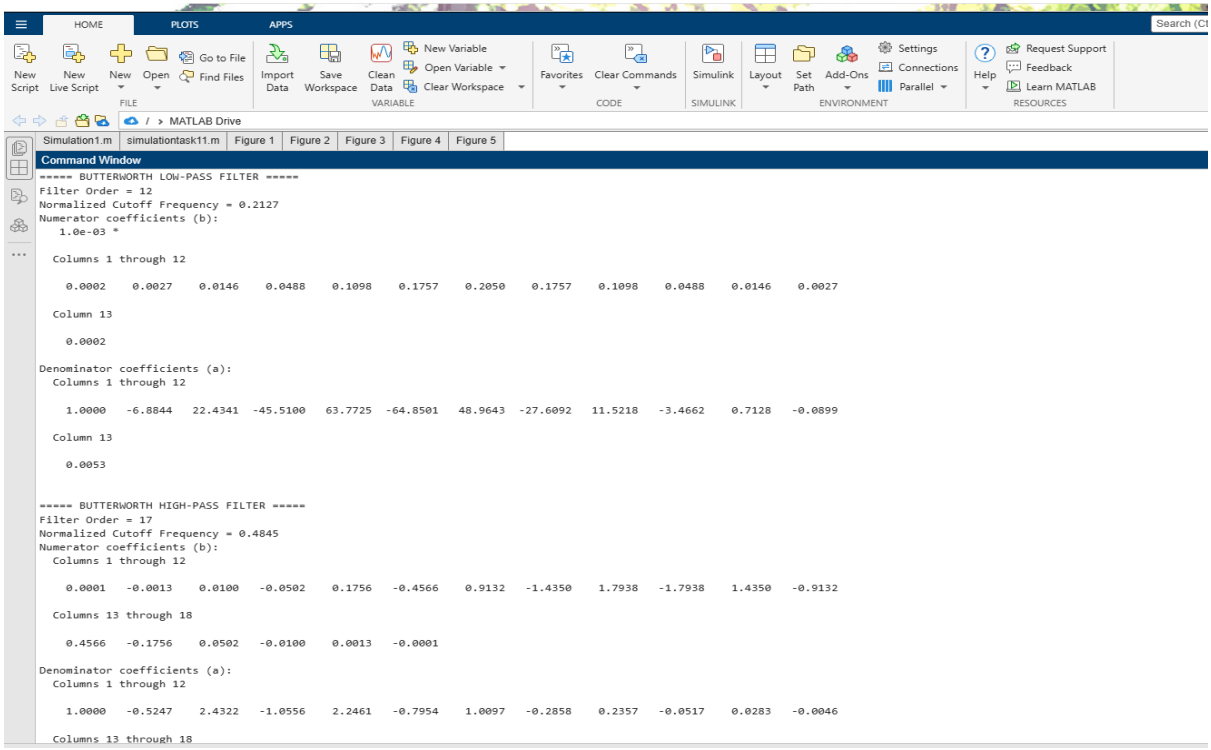

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Krishna Sri Reddy
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Simulation1.m
simulationtask11.m
Workspace
Editor
Simulation1.m
simulationtask11.m
/MATLAB Drive/Simulation1.m
1 clc;
2 clear;
3 close all;
4 %% Input Specifications
5 Fs = 1000; % Sampling frequency (Hz)
6 % Low-pass specifications
7 Fp_LP = 100; % Passband edge (Hz)
8 Fst_LP = 150; % Stopband edge (Hz)
9 % High-pass specifications
10 Fst_HP = 200; % Stopband edge (Hz)
11 Fp_HP = 250; % Passband edge (Hz)
12 % Attenuation specifications
13 Ap = 1; % Passband attenuation (dB)
14 Ast = 40; % Stopband attenuation (dB)
Apps
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Butterworth Low-Pass Filter - Magnitude Response
Magnitude (dB)
0
-200
-400
0
100
200
300
400
500
Frequency (Hz)
Butterworth Low-Pass Filter - Phase Response
Phase (radians)
0
-10
-20
-30
0
100
200
300
400
500
Frequency (Hz)


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Simulation1.m
simulationtask11.m
Figure 1
Figure 2
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Figure 5
Command Window
===== BUTTERWORTH LOW-PASS FILTER =====
Filter Order = 12
Normalized Cutoff Frequency = 0.2127
Numerator coefficients (b):
1.0e-03 *
Columns 1 through 12
0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027
Column 13
0.0002
Denominator coefficients (a):
Columns 1 through 12
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899
Column 13
0.0053
===== BUTTERWORTH HIGH-PASS FILTER =====
Filter Order = 17
Normalized Cutoff Frequency = 0.4845
Numerator coefficients (b):
Columns 1 through 12
0.0001 -0.0013 0.0100 -0.0502 0.1756 -0.4566 0.9132 -1.4350 1.7938 -1.7938 1.4350 -0.9132
Columns 13 through 18
0.4566 -0.1756 0.0502 -0.0100 0.0013 -0.0001
Denominator coefficients (a):
Columns 1 through 12
1.0000 -0.5247 2.4322 -1.0556 2.2461 -0.7954 1.0097 -0.2858 0.2357 -0.0517 0.0283 -0.0046
Columns 13 through 18